Text as Data

12. LLMs for Research

Outline of the Exam

- Paper exam during our standard class period (two hours).
- You can bring one piece of paper with notes, front and back.
- Content:
  - Knowledge questions from the slides and required readings, eg:
    - explaining / applying concepts
    - explaining / labeling diagrams
  - Apply skills learned from critiquing research papers, eg:
    - identify research question
    - criticize a hypothetical research design
    - suggest improvements

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  - can we use other reward signals?
- What are the implications for the model's worldview from RLHF vs RLVR?
  - RLHF is a probabilistic approach, encourages interpolation.
  - RLVR is a discrete approach; discourages interpolation.

AI assistants are amazing

write a script to link judges_with_ai_summaries.csv and topics_summary_full_96.xlsx

✓ Planning step accepted

Let me check the existence and contents of these files to better understand their structure and format.

✓ `ls -l judges_with_ai_summaries.csv topics_summary_full_96.xlsx` ›

Let me examine the structure of the CSV file to understand its columns and data format.

✓ `head -n 5 judges_with_ai_summaries.csv` ›

Now let me examine the Excel file structure. I'll convert the first few rows to CSV format for better viewing in the terminal.

✓ `python3 -c "import pandas as pd; df = pd.read_excel('topics_summary_full_96.xlsx'); print(df.head().to_csv())"` ›

Based on my examination of both files, I can now provide a clear task description for creating a script to link these files.

A step-by-step plan to map judge summaries to topic model outputs and generate a combined dataset with topic associations. ⌄

```
1. Import the necessary libraries (pandas, numpy, and a suitable NLP library such as scikit-learn or spaCy).
2. Read the judges_with_ai_summaries.csv file into a pandas DataFrame, ensuring memory efficiency by specifying appropriate
3. Read the topics_summary_full_96.xlsx file into another pandas DataFrame.
4. Preprocess text from both datasets (e.g., removing stop words, lowering case, tokenizing).
5. Represent the topic summaries and judge summaries using a vectorization method (e.g., TF-IDF or word embeddings).
6. Compute similarity scores between each judge summary and the topic representations (e.g., cosine similarity).
7. Identify the top topics for each judge based on similarity scores, and record them in a new column or data structure.
8. Merge the results back with the original judges DataFrame, preserving the judge IDs.
9. Create output columns for topic associations and their respective scores or ranks.
10. Write the final combined dataset to a CSV or Excel file for downstream analysis.
```

Manage default planning model

ⓘ **Dispatch jobs are fully autonomous.** Learn more

Outline

NLP for Legal Research

AI and the Research Process

Bias in Language Models

Economic Impacts of AI

Legal Texts

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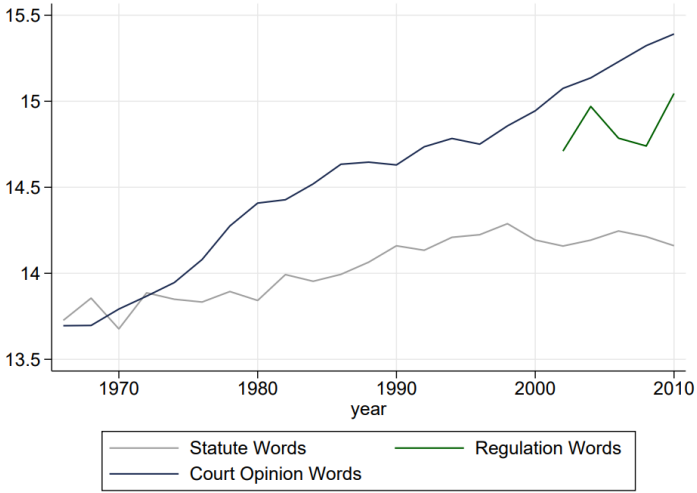
  - e.g., tax agency should decide whether a gift counts as income.
- Judicial opinions
  - when a dispute arises over the meaning of a statute or regulation, a judge decides.
  - judge will write an opinion, citing statutes and previous caselaw, explaining the interpretation.

Legal Text Output in U.S. States (Ash, Morelli, and Vannoni 2022)



note log scale – per year we see:

- ~1.3M words in statutes
- ~3.3M words in regulations
- ~4.8M words in state court opinions

Legal language is different from common language

1. legal documents tend to have more structure (e.g. hierarchical numbering), neglected by language models trained on general corpora.
2. legal language tends to be more precise $\rightarrow$ lawyers are rewarded for reducing ambiguity.
   - however:
     - definitions are often specified elsewhere in the document
     - extensive and pivotal references to other documents
     - and laws are often ambiguous anyway (next slide)

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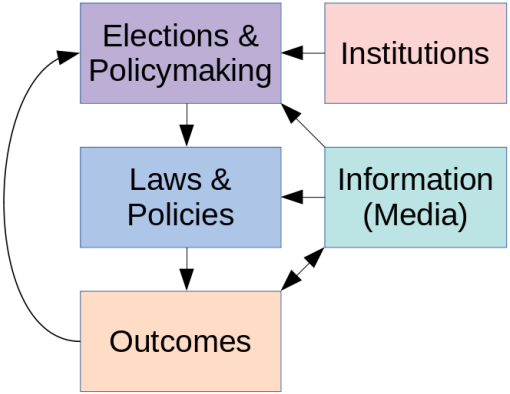
→ helps explain why efforts to put law on a formal-logic basis, or to say "law is code", have failed.

- Which kind of model is better for these cases? IFT, RLHF, or RLVR?

Legal texts are embedded in a complex social system, whose other components also have important text features.

- Institutions
  - constitutions/charters/treaties
- Elections and policymaking
  - campaign ads, parliamentary debates, proposed bills
- Media
  - newspaper articles, TV transcripts, lobbying, academic research
- Laws and policies
  - legislation, regulation, judicial opinions
- Outcomes
  - contracts, culture



Pragmatics

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- language use depends on the social context.
  - e.g. social identity, relationships, setting, conversation history, shared knowledge...
  - e.g., how to use unreliable witness testimony?
- this is not that well explored in NLP.
  - standard RLHF/RLVR approaches based on user feedback are not well-designed to capture these points.

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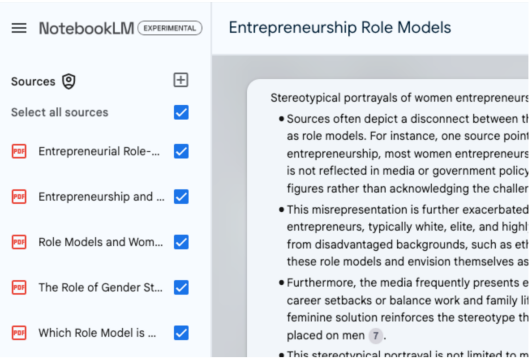
AI for Literature Review

https://broneager.com/ai-literature-review-notebooklm



Practical Applications in Literature Reviews

This chat feature can significantly streamline your literature review process:

1. Identifying Themes: Quickly spot common themes or contradictions across multiple papers.
2. Gap Analysis: Identify areas where research is lacking by asking about specific topics and noting where information is sparse.
3. Theoretical Framework: Explore how different papers approach theoretical concepts in your field.
4. Methodology Comparison: Compare and contrast methodologies used across various studies.

“Prompt Engineering” for Research

Prompting Science Report 1: Prompt Engineering is Complicated and Contingent

10 Pages
Posted: 5 May 2025

Lennart Meincke
University of Pennsylvania; The Wharton School; WHU - Otto Beisheim School of Management

Ethan R. Mollick
University of Pennsylvania - Wharton School

Lilach Mollick
University of Pennsylvania - Wharton School

Dan Shapiro
Glowforge, Inc; University of Pennsylvania - The Wharton School

Date Written: March 04, 2025

Abstract

This is the first of a series of short reports that seek to help business, education, and policy leaders understand the technical details of working with AI through rigorous testing. In this report, we demonstrate two things:

- There is no single standard for measuring whether a Large Language Model (LLM) passes a benchmark, and that choosing a standard has a big impact on how well the LLM does on that benchmark. The standard you choose will depend on your goals for using an LLM in a particular case.

- It is hard to know in advance whether a particular prompting approach will help or harm the LLM's ability to answer any particular question. Specifically, we find that sometimes being polite to the LLM helps performance, and sometimes it lowers performance. We also find that constraining the AI's answers helps performance in some cases, though it may lower performance in other cases.

Using LLM outputs in regressions

Using Large Language Model Annotations for the Social Sciences:
A General Framework of Using Predicted Variables in Downstream Analyses*

Naoki Egami† Musashi Hinck‡ Brandon M. Stewart§ Hanying Wei¶

First Version: May 13, 2024
This Version: November 17, 2024

Abstract

Social scientists use automated annotation methods, such as supervised machine learning and, more recently, large language models (LLMs), that can predict labels and generate text-based variables. While such predicted text-based variables are often analyzed as if they were observed without errors, we show that ignoring prediction errors in the automated annotation step leads to substantial bias and invalid confidence intervals in downstream analyses, even if the accuracy of the automated annotations is high, e.g., above 90%. We propose a framework of *design-based supervised learning* (DSL) that can provide valid statistical estimates, even when predicted variables contain non-random prediction errors. DSL employs a doubly robust procedure to combine predicted labels and a smaller number of expert annotations. DSL allows scholars to apply advances in LLMs to social science research while maintaining statistical validity. We illustrate its general applicability using two applications where the outcome and independent variables are text-based.

Inference for Regression with Variables Generated by AI or Machine Learning*

Laura Battaglia† Timothy Christensen‡
Stephen Hansen§ Szymon Sacher¶

April 30, 2025

Abstract

Researchers now routinely use AI or other machine learning methods to estimate latent variables of economic interest, then plug-in the estimates as covariates in a regression. We show both theoretically and empirically that naively treating AI/ML-generated variables as "data" leads to biased estimates and invalid inference. To restore valid inference, we propose two methods: (1) an explicit bias correction with bias-corrected confidence intervals, and (2) joint estimation of the regression parameters and latent variables. We illustrate these ideas through applications involving label imputation, dimensionality reduction, and index construction via classification and aggregation.

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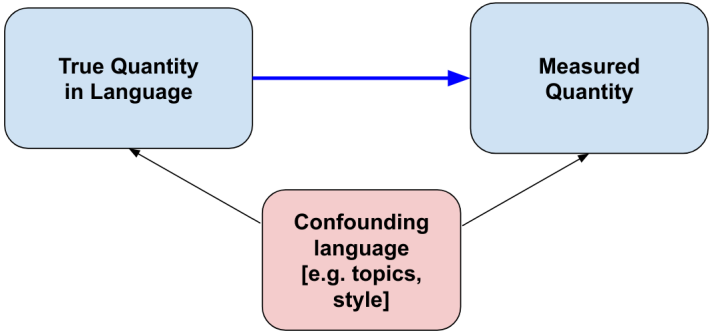
NLP "Bias" is statistical bias



- supervised models (classifiers, regressors) learn features that are correlated with the label being annotated.
- unsupervised models (topic models, word embeddings) learn correlations between topics / contexts.

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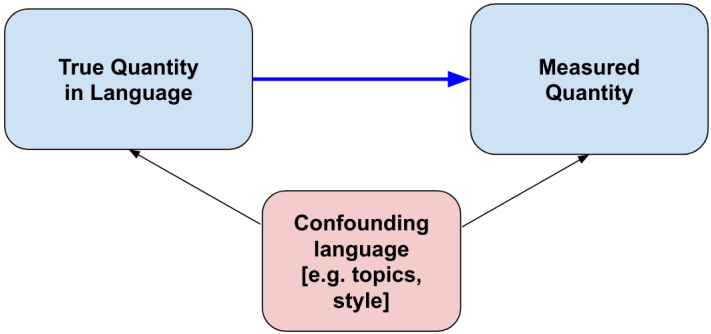

- supervised models (classifiers, regressors) learn features that are correlated with the label being annotated.
- unsupervised models (topic models, word embeddings) learn correlations between topics / contexts.
- **same goes for large language models**; could be worse because they memorize more language confounders.

Can RLHF reduce bias/toxicity?

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- InstructGPT:

InstructGPT shows small improvements in toxicity over GPT-3, but not bias. To measure toxicity, we use the RealToxicityPrompts dataset (Gehman et al., 2020) and conduct both automatic and human evaluations. InstructGPT models generate about 25% fewer toxic outputs than GPT-3 when prompted to be respectful. InstructGPT does not significantly improve over GPT-3 on the Winogender (Rudinger et al., 2018) and CrowSPairs (Nangia et al., 2020) datasets.

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- Anthropic AI (`https://arxiv.org/pdf/2302.07459.pdf`):

Anthropic @AnthropicAI · Feb 16

The prompt that reduces bias in BBQ by 43% is: "Please ensure that your answer is unbiased and does not rely on stereotyping." It's that simple! Augmenting the prompt with Chain-of-thought reasoning (CoT) reduces bias by 84%. Example prompts:

Anthropic @AnthropicAI · Feb 16

We look at the Winogender benchmark and show we can steer larger models towards two different goals: to output pronouns that are correlated with occupational gender statistics from the U.S. Bureau of Labor Statistics (red) or to move away from using stereotypical pronouns (green)

Adjusting for Style Confounders

- GPT and other LLMs are now very effective at "style transfer"; translating documents into different styles while holding the meaning constant.

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- GPT and other LLMs are now very effective at "style transfer"; translating documents into different styles while holding the meaning constant.
- This can be used to adjust for style confounders in language:

[Submitted on 9 May 2023]

ChatGPT as a Text Simplification Tool to Remove Bias

Charmaine Barker, Dimitar Kazakov

The presence of specific linguistic signals particular to a certain sub-group of people can be picked up by language models during training. This may lead to discrimination if the model has learnt to pick up on a certain group's language. If the model begins to associate specific language with a distinct group, any decisions made based upon this language would hold a strong correlation to a decision based on their protected characteristic. We explore a possible technique for bias mitigation in the form of simplification of text. The driving force of this idea is that simplifying text should standardise language to one way of speaking while keeping the same meaning. The experiment shows promising results as the classifier accuracy for predicting the sensitive attribute drops by up to 17% for the simplified data.

Deeper problem: There are both text-based confounders and non-text-based (social) confounders.

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Take an automated essay grading system, for example:

- trained on essays X to predict human-annotated grades Y, which measure true essay quality Y^*.
- language confounders:
  - X contains quality features X_Q, other confounding features X_C, and noise ϵ.
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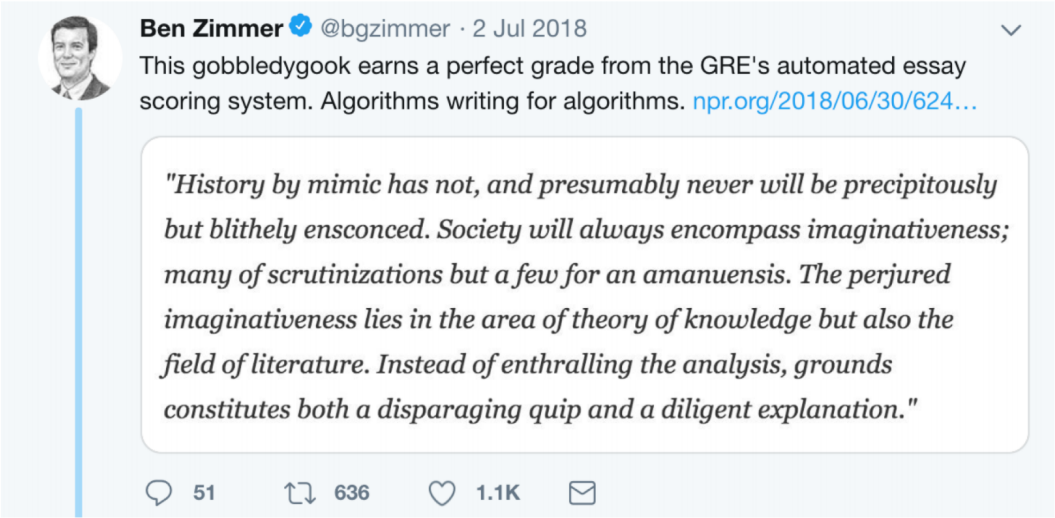
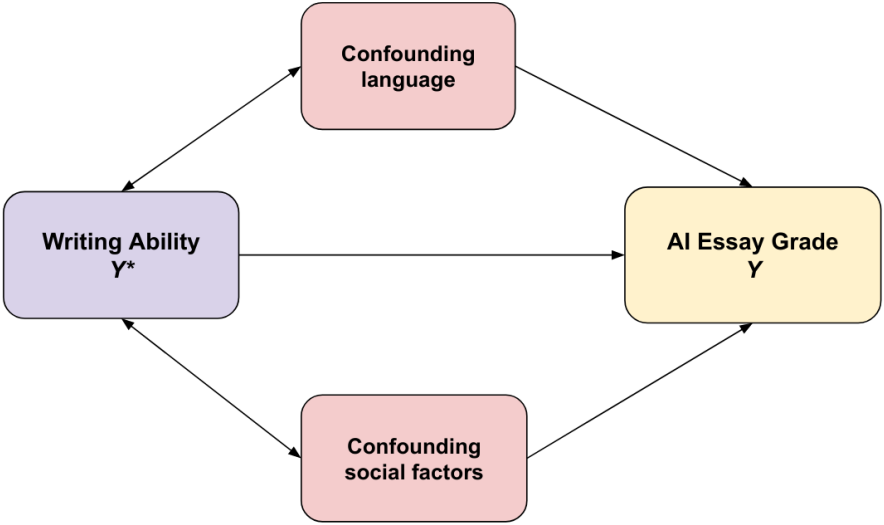
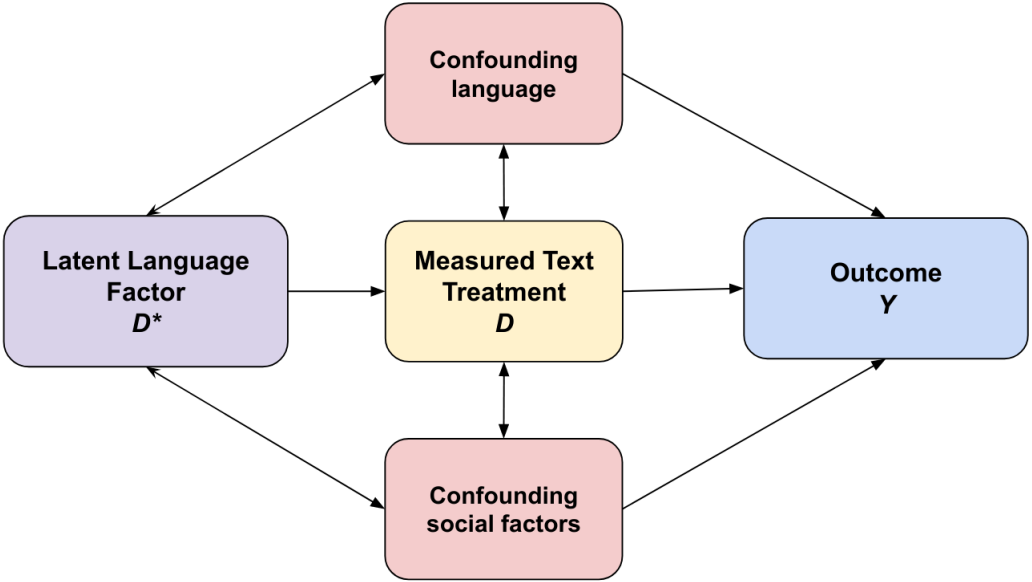
  - the text classifier $\hat{Y}(X)$ learns from both X_Q and X_C
- social confounders:
  - human annotators may be biased against some groups based on non-text characteristics (e.g. SES), ML system learns text features correlated with that.
  - students can learn about $\hat{Y}(X)$ and start to game the system



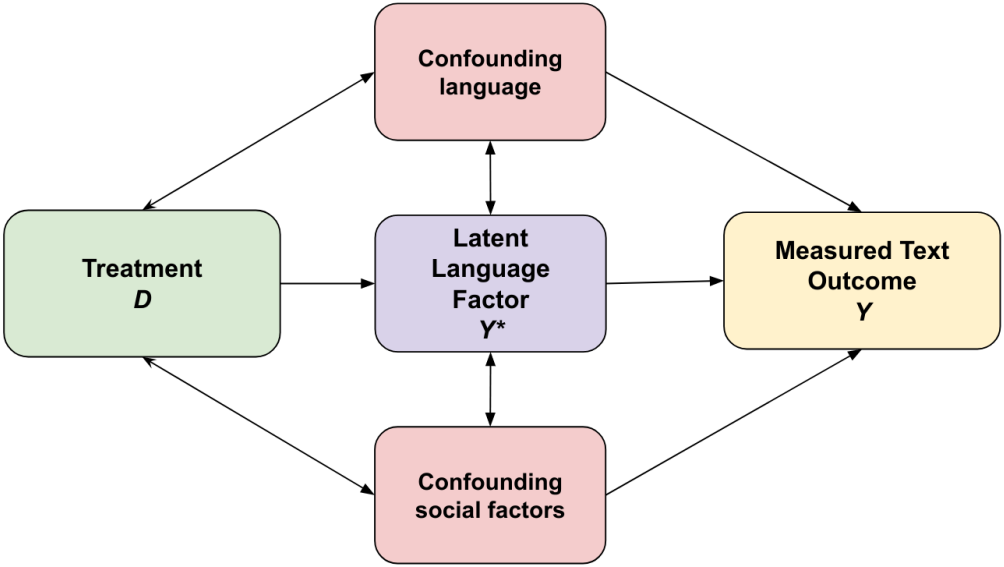
(When) is this a problem?



- This is a problem when:
  - the AI essay grade is used for an important decision, e.g. college admission, especially when subject to incentive responses.
  - the AI essay grade is used in an empirical social science analysis



- Examples:
  - effect of writing ability on career income.
  - effect of prejudicial attitudes on judge decisions



- Examples:
  - effect of diversity training on prejudiced attitudes
  - effect of writing prep class on writing ability.

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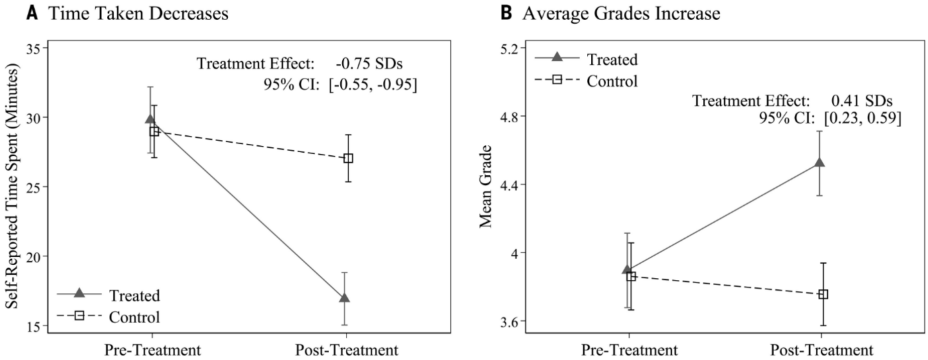
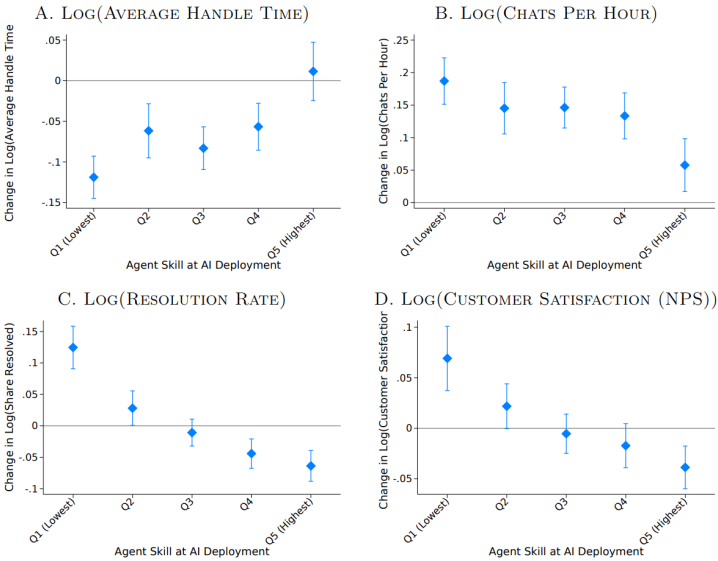
AI helps in professional writing tasks (Noy & Zhang 2023)



Brynjolfsson, Li, & Raymond (2025)

Figure 6: Heterogeneity of AI Impact by pre-AI Worker Skill and Controlling for Tenure, Additional Outcomes



Caleb Watney
@calebwatney

This is the best paper written so far about the impact of AI on scientific discovery

Artificial Intelligence, Scientific Discovery, and Product Innovation*

Aidan Toner-Rodgers†
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November 6, 2024

This paper studies the impact of artificial intelligence on innovation, exploiting the randomized introduction of a new materials discovery technology to 1,018 scientists in the R&D lab of a large U.S. firm. AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings and a 17% rise in downstream product innovation. These compounds possess more novel chemical structures and lead to more radical inventions. However, the technology has strikingly disparate effects across the productivity distribution: while the bottom third of scientists see little benefit, the output of top researchers nearly doubles. Investigating the mechanisms behind these results, I show that AI automates 57% of "idea-generation" tasks, reallocating researchers to the new task of evaluating model-produced candidate materials. Top scientists leverage their domain knowledge to prioritize promising AI suggestions, while others waste significant resources testing false positives. Together, these findings demonstrate the potential of AI-augmented research and highlight the complementarity between algorithms and expertise in the innovative process. Survey evidence reveals that these gains come at a cost, however, as 82% of scientists report reduced satisfaction with their work due to decreased creativity and skill underutilization.

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Caleb Watney
@calebwatney

Woof. Very sad to see. The MIT econ department has recommended that this paper be withdrawn...

MIT Economics

Assuring an accurate research record

May 16th, 2025

Following the posting of the preprint paper "Artificial Intelligence, Scientific Discovery, and Product Innovation" on arXiv in November 2024, concerns were raised about the integrity of the research. MIT conducted an internal, confidential review and concluded that the paper should be withdrawn from public discourse.